

Matias Turkulainen

PROFILE	Computer vision PhD candidate at Aalto University, advised by Juho Kannala and Arno Solin . My research has focused on 3D Reconstruction, Gaussian Splatting, and Neural Rendering.	
EDUCATION	Aalto University · Espoo, Finland	2024 – Dec 2028
	<i>Ph.D., Computer Science</i>	
	ETH Zürich · Zürich, Switzerland	2021 – 2024
	<i>M.Sc., Robotics, Systems and Control</i>	
	University of Glasgow · Glasgow, Scotland	2017 – 2021
	<i>BEng (Hons), Mechanical Engineering with Aeronautics</i>	
EXPERIENCE	Niantic Spatial · London, UK	Summer 2025 and Summer 2026
	<i>Research Intern</i>	
	Aalto University	Dec 2023 – May 2024
	<i>Research Assistant</i>	
	VTT, Technical Research Centre of Finland	Apr 2023 – Nov 2023
	<i>Research Intern</i>	
	VLG, ETH Zürich	Feb 2022 – Jul 2022
	<i>Research Assistant</i>	
	Nokia	Jun 2020 – Aug 2021
	<i>Drone Systems Trainee</i>	
SELECTED PUBLICATIONS	TileGS: Tile-Local Depth Binning for Gaussian Splatting Rasterization	PG 2026
	<i>Wei Tan, Matias Turkulainen, Lauri Ilola, Hamed Rezazadegan Tavakoli, Juho Kannala.</i>	
	Cross-View Splatter: Feed-Forward View Synthesis with Georeferenced Images	CVPR 2026
	<i>Matias Turkulainen, Akshay Krishnan, Filippo Aleotti, Mohamed Sayed, Guillermo Garcia-Hernando, Juho Kannala, Arno Solin, Gabriel Brostow, Daniyar Turmukhambetov.</i>	
	DeSplat: Decomposed Gaussian Splatting for Distractor-Free Rendering	CVPR 2025
	<i>Yihao Wang, Marcus Klasson, Matias Turkulainen, Shuzhe Wang, Juho Kannala, Arno Solin.</i>	
	AGS-Mesh: Adaptive Gaussian Splatting and Meshing with Geometric Priors for Indoor Room Reconstruction Using Smartphones	3DV 2025
	<i>Xuqian Ren, Matias Turkulainen, Jiepeng Wang, Otto Seiskari, Iaroslav Melekhov, Esa Rahtu, Juho Kannala.</i>	
	DN-Splatter: Depth and Normal Priors for Gaussian Splatting and Meshing	WACV 2025
	<i>Matias Turkulainen, Xuqian Ren, Iaroslav Melekhov, Otto Seiskari, Juho Kannala, Esa Rahtu.</i>	
OPEN SOURCE	Gaussian Splatting on the Move: Blur and Rolling Shutter Compensation for Natural Camera Motion	ECCV 2024
	<i>Otto Seiskari, Jerry Ylilammi, Valtteri Kaatrasalo, Pekka Rantalankila, Matias Turkulainen, Juho Kannala, Esa Rahtu, Arno Solin.</i>	
	gsplat: An Open-Source Library for Gaussian Splatting	JMLR 2025
	<i>Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, Angjoo Kanazawa.</i>	
	gsplat	since 2023
	<i>Core developer · modular library for 3D Gaussian Splatting</i>	
	Nerfstudio	since 2023
	<i>Maintainer · collaborative framework for neural rendering</i>	
INVITED TALKS & EVENTS	Google PhD Summit · Zürich	Jun 2026
	FAI SIG in Computer Vision Workshop	Dec 2024
	FMX 2024 · Stuttgart, Germany	Apr 2024
	Machine Learning Coffee Seminar · FAI	May 2024
	Gaussian Splatting Online Meetup	Jul 2024